

Arnav Kakkar

+91 9996438080 arnavkakk28.3@gmail.com
arnav-kakkar arnavkakk28

Education

Thapar Institute of Engineering and Technology, Patiala	Sep 2023 – Jun 2027
B.E. Electronics Engineering	
The Vivekanand School, New Delhi	Apr 2021 – Jun 2023
CBSE Senior Secondary	

Experience

Software Development Engineer (SDE) Intern	Feb 2026 – Mar 2026
Bluestock Fintech — Remote	
- Completed technical assessment with a score of 100/100 and contributed to React, Node.js, and Firebase-based applications with REST API integration, improving responsiveness and usability. - Developed UI components using React and Tailwind CSS while supporting feature implementation and workflow optimization in a Git-based environment, ensuring clean code and consistent interface design.	

Projects

RAG-Based AI Teaching Assistant	Jan 2026 – Feb 2026
Python, LangChain, FAISS, OpenAI API, NLP	
- Built a Retrieval-Augmented Generation (RAG) system using LangChain and FAISS to query 500+ documents, enabling fast and context-aware responses across 50+ queries per session. - Developed an embedding-based semantic search pipeline, increasing answer relevance by 30% and reducing incorrect outputs by 25% through improved context retrieval. - Optimized retrieval and response workflow, reducing query latency by 35% while supporting concurrent users with stable and efficient performance.	
Real-Time Anomaly Detection System	Apr 2026 – May 2026
Python, Scikit-learn, Streamlit, Plotly	
- Built a real-time anomaly detection system using Isolation Forest on 1000+ streaming data points with continuous tracking, monitoring, and detection of unusual patterns in data. - Created 5+ time-series features and combined ML with rule-based logic, improving anomaly detection accuracy by 20% across different test cases and varying data conditions. - Developed a Streamlit dashboard to show 100+ live data points and generate 3–4 clear explanations per anomaly, helping users better understand patterns and take decisions.	

Leadership

Executive Core Member	Feb 2024 – Jul 2025
Frosh (under the aegis of Dean of Student Affairs)	Thapar Institute of Engineering and Technology, Patiala
- Led 79 mentors as part of 20-member core team handling 20,000+ applicants. - Coordinated 30-day Fresher's Fest with 3,200+ participants. - Headed Media Department for event coverage and coordination.	
Executive Committee Member — Saturnalia Tech Fest	Sep 2025 – Nov 2025
- Led media operations for 400+ member team. - Increased inter-college participation through outreach. - Executed events with 3000+ attendees.	

Certifications

The Ultimate Job Ready Data Science Course	Nov 2025 - Mar 2026
codewithharry.com	
Deloitte Data Analytics Job Simulation	May 2026 - May 2026
theforage.com	

Skills

Languages: Python, JavaScript, C, SQL
Frameworks/Tools: React.js, Firebase, Pandas, NumPy, Matplotlib, Git, GitHub, Vercel, Jupyter Notebook
Core Skills: Data Structures, Algorithms, REST APIs, Authentication Systems, Full-Stack Development, Data Analysis, Debugging
AI/ML: RAG, LLM Integration, Semantic Search, Data Preprocessing, EDA